PS No.	**78**
Project Title	**Blood Bank Management System**
Problem Statement	Build a program for blood-bank staff to add donors, issue blood, view stock and search donors by blood group. Keep stock in a dictionary and donors in a list saved to donors.txt. Issue blood only if enough units are available. **Language:** Python 3 **Concepts:** Dictionaries, Lists, File Handling, Search **Note:** Initial stock: A+ 10, B+ 5, O+ 8, AB+ 2, all other groups 0. **Note:** donors.txt for Test Case 4 contains: Meena Joshi, O+, 1 and Suresh Rao, O+, 2. **Note:** Each test case starts from this initial data.
Test Cases / Case Statements (4 cases)	

INPUT	EXPECTED OUTPUT
Test Case 1: Normal Input	
Menu: Add donor Name: Anil Verma Blood group: B+ Units: 1	Donor saved to donors.txt. Stock of B+ is now 6 units.
Test Case 2: Boundary / Minimum Input	
Menu: Issue blood Blood group: AB+ Units: 2	2 unit(s) of AB+ issued. Stock of AB+ is now 0 units.
Test Case 3: Invalid Input	
(a) Menu: Issue blood, group A-, units 3 (b) Menu: Add donor, name Rina Das, group C+, units 1	(a) Insufficient stock: only 0 unit(s) of A- available. (b) Invalid blood group. Valid groups: A+, A-, B+, B-, AB+, AB-, O+, O-.
Test Case 4: Different Valid Scenario	
Menu: Search donors Blood group: O+	Donors with blood group O+: 1. Meena Joshi - 1 unit(s) 2. Suresh Rao - 2 unit(s) Total donors found: 2

Bare Minimum Requirements	**1.** Write the solution in Python 3 (one .py file) and run it from the console. **2.** Show a menu (Add donor, Issue blood, View stock, Search donors, Exit). **3.** Keep blood stock in a dictionary and donor records in a list of dictionaries. **4.** Accept only the eight valid blood groups and positive whole-number units. **5.** Test the program with all four test cases.
Instruction	**1.** Understand the problem statement and define the problem clearly. **2.** Develop the solution using proper logic and programming concepts. **3.** Implement all required features and pass the given test cases. **4.** Test and debug the project to ensure correct output. **5.** Prepare the documentation/report and upload the code and report to GitHub. **6.** Complete and submit the project within the given deadline.